

ADDIS ABABA UNIVERSITY

ADDIS ABABA INSTITUTE OF TECHNOLOGY

School of Mechanical and Industrial Engineering (SMIE)

Course Title: Engineering Mechanics-II (Dynamics)

A/Year: 2021/22

Course No: MEng 2052

Semester: II

Credit Hrs: 3

ECTS: 5

Prerequisite: Engineering Mechanics-I and Applied Mathematics-I

Course-outline

Chapter-1: Introduction ----- ----- (1 week)

- 1.1. Basic Concepts
- 1.2. System of Units
- 1.3. Newton's Law of Motions
- 1.4. Gravitation

Chapter-2: Kinematics of Particles --- ----- (5 weeks)

- 2.1. Rectilinear Motion
- 2.2. Plane curve linear Motion
- 2.3. Rectangular Coordinates
- 2.4. Normal Tangential Coordinates
- 2.5. Polar Coordinates
- 2.6. Relative Motion (translating Axes)
- 2.7. Constrained Motion

Chapter-3: Kinetics of particles ----- ----- (4 weeks)

- 3.1. Force, Mass, acceleration
- 3.2. Newton's second Law
- 3.3. Rectilinear Motion
- 3.4. Curvilinear motion
- 3.5. Work and energy
- 3.6. Work and kinetic Energy
- 3.7. Potential Energy
- 3.8. Impulses and Momentum
- 3.9. Linear impulse and Linear Momentum
- 3.10. Angular Impulse and Angular Momentum
- 3.11. Impact

Chapter-4: Plane Kinematics of Rigid Bodies ----- (2 weeks)

- 4.1. Rotation
- 4.2. Absolute Motion
- 4.4. Relative velocity and Acceleration
- 4.5. Instantaneous Center of Zero Velocity
- 4.6. Motion of Relative Rotating Axes

Chapter-5: Plane kinetics of Rigid Bodies ----- (2 weeks)

- 5.1. Force, Mass and Acceleration
- 5.2. General Equation of Motion
- 5.3. Translation
- 5.4. Fixed Axis Rotation
- 5.5. General Plane motion
- 5.6. Work and Energy Relations
- 5.7. Impulse and Momentum Equations

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Text: Engineering Mechanics: Dynamics, **J.L. Meriam, John wiley &sons, Inc**

Reference:

1. Engineering Mechanics: An Introduction of Dynamic, **David J. McGill and Wilton**
2. Vector Mechanics of Engineering by **F.P. Beer and E.R.Johanson**
3. Engineering Mechanics by **R.C. Hibbeler**
4. Statics and Dynamics by **J.H.Ginsberg**

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Evaluation

Assignment	15%
Quiz	10%
Intermediate Exam	30%
Final Exam	45%

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Grading system: SL-2013: Grading rules for undergraduate program.

NB: It is mandatory students to attend class minimum of **75%** attendance.